

GateIO: A Yaw-Rate-Gated Causal TCN for UAV GPS-Outage Bridging

Nandini Saxena

Bharati Vidyapeeth, Pune

nandinisaxenawork@gmail.com

Preprint — submitted to arXiv, 2025

Abstract

GPS outages are a practical failure mode for UAVs operating in degraded signal environments. When an outage coincides with a coordinated manoeuvre, constant-velocity extrapolation accumulates large errors because horizontal velocity changes systematically with yaw rate. We present GateIO, a causal temporal model that predicts GPS velocity during outages from inertial measurements and the last available GPS-aided velocity, while applying a yaw-rate-gated velocity-persistence loss during low-yaw-rate windows. On a within-flight validation split of five MARS-LVIG flight sequences (Li et al., 2024), GateIO reduces mean endpoint drift on six turning sequences from 135.79 m (constant-velocity baseline) to 4.73 m, with an overall mean of 6.72 m across 59 sequences. Removing the temporal convolutional backbone increases drift by $43.97\times$ under the gated loss, compared to approximately $1\times$ under a prediction-only objective, which we interpret as the gated prior encouraging the network to use temporal context. GateIO performs substantially worse than constant-velocity on a false-alarm group — 28.08 m versus 4.68 m — where motor vibration produces accelerometer patterns similar to turning manoeuvres. Performance also degrades substantially on the test split due to heading-distribution bias in the within-flight split design. The prior is motivated by the relationship between yaw rate and horizontal velocity change in approximately coordinated flight; we evaluate it on a multirotor platform (DJI M300 RTK) as a data-driven motion prior rather than as a complete aircraft model. A bag-level evaluation split and an explicit velocity-increment prediction target are being developed to assess and potentially improve cross-flight generalisation. Code and models are available at <https://github.com/nandini1612/gateio>.

1. Introduction

UAVs operating in GPS-degraded environments may need to bridge outages lasting several seconds to tens of seconds without new satellite measurements. In our experiments, the simulated outage lasts 10 seconds within a 30-second sequence. At typical cruise speeds of 3 to 12 m/s, 10 seconds of flight covers 30 to 120 m of ground track. The standard fallback — holding the last known velocity constant until GPS returns — accumulates large errors when a manoeuvre coincides with the outage, because horizontal velocity changes systematically with yaw rate even when forward speed remains approximately constant.

A common approach is a Kalman filter with a constant-velocity or constant-acceleration process model. A well-tuned filter reduces drift substantially on straight-dominated flight, but its process model has no representation of the systematic relationship between yaw rate and horizontal velocity change during a manoeuvre. Adapting the filter's noise covariances (Brossard et al., 2020) improves accuracy when the

process model is approximately correct, but does not resolve the structural mismatch during manoeuvring flight.

Learned inertial odometry offers an alternative. Chen et al. (2018) showed that a recurrent network trained on windowed IMU data can estimate pedestrian displacement as a polar vector — travelled distance and heading change — without per-subject calibration. The paper also evaluated the approach on non-periodic motion, including shopping trolleys and baby strollers. Herath et al. (2020) scaled this to the RoNIN benchmark with three backbone variants, finding that a temporal convolutional network matched recurrent accuracy while training faster. Liu et al. (2020) combined a learned displacement predictor with a stochastic cloning EKF in TLIO. More recent aerial extensions include DPI-LSTM (Liang et al., 2026), which incorporates selected strapdown inertial-navigation relationships into differentiable loss terms for quadrotor pose estimation, and AirIO (Qiu et al., 2025), which addresses body-frame IMU feature observability and predicts velocity uncertainty for EKF fusion.

GateIO replaces the filter entirely during GPS outages and encodes a single physically motivated prior: during windows where yaw rate is low, horizontal velocity should remain close to its last known value. This relationship is particularly natural for approximately coordinated fixed-wing cruise, but we evaluate it here on a multirotor dataset as a data-driven motion prior rather than as a complete aircraft model. The model predicts GPS velocity in normalised space; the gated loss targets the last known GPS velocity on straight windows, approximating a velocity-persistence constraint without requiring an absolute-speed parameter. The model operates without new GPS measurements during the outage but uses IMU data together with the last available GPS-aided velocity.

We evaluate on the MARS-LVIG dataset (Li et al., 2024), a UAV dataset released for SLAM evaluation. The platform is a DJI M300 RTK multirotor. To the best of our knowledge, no prior published work has used its IMU and GPS channels for GPS-outage bridging; we searched Google Scholar and Semantic Scholar for "MARS-LVIG" combined with "dead reckoning," "inertial navigation," and "velocity prediction" through December 2025 and found no relevant prior results. We emphasise that all current results are from a within-flight chronological split and do not assess cross-flight generalisation.

The contributions of this paper are: *(i)* GateIO, a yaw-rate-gated causal TCN that reduces turning-sequence mean drift from 135.79 m to 4.73 m on the current validation split; *(ii)* a yaw-rate-gated velocity-persistence regulariser applied to UAV GPS-outage bridging; *(iii)* the observation that the TCN backbone shows $43.97\times$ ablation sensitivity under the gated loss but approximately $1\times$ under a prediction-only objective; *(iv)* a quantitative analysis of the Kalman filter's turning-sequence failure consistent with pre-outage lateral acceleration absorbed into the bias estimate; and *(v)* an honest characterisation of current limitations including the false-alarm group failure, within-flight split design, platform mismatch, and single-run results.

2. Related Work

2.1 Learned Inertial Odometry

Chen et al. (2018) proposed IONet, formulating pedestrian inertial odometry as inferring a polar displacement vector — travelled distance and heading change — from fixed-length IMU segments using a recurrent network. The method avoided per-subject step-detector calibration and was also evaluated on non-periodic motion including shopping trolleys and baby strollers. Herath et al. (2020) scaled learned inertial odometry to the RoNIN benchmark with over 100 subjects and three backbone variants (ResNet, LSTM, temporal convolutional network), finding that the TCN variant matched recurrent accuracy while training faster. Liu et al. (2020) integrated a learned displacement predictor with a stochastic cloning EKF in TLIO, using the network's predicted uncertainty as measurement covariance. Rao et al. (2022) proposed CTIN, combining a ResNet feature extractor with both local and global multi-head self-attention, and demonstrated improved accuracy on multiple pedestrian inertial odometry benchmarks. All of these methods target pedestrian motion and benefit from gait regularity that does not apply to UAV flight.

Qiu et al. (2025) propose AirIO for aerial inertial odometry, training in the body frame and predicting velocity uncertainty for EKF fusion. Cioffi et al. (2023) apply learned inertial odometry to agile quadrotor racing. Liang et al. (2026) incorporate selected strapdown inertial-navigation relationships into differentiable loss terms for quadrotor pose estimation in DPI-LSTM. GateIO differs from these aerial methods in targeting a temporary GPS outage where the pre-outage velocity is available and encoding a single-regime motion prior rather than full strapdown kinematics.

2.2 Classical and Hybrid Dead Reckoning

Brossard et al. (2020) train a dilated convolutional network to predict process and measurement noise covariances for an invariant EKF, achieving competitive results on KITTI using IMU alone. The design retains the filter structure and cannot compensate for a structural mismatch between the constant-velocity process model and manoeuvring dynamics. We implement a six-state Kalman filter with quaternion-based gravity compensation and validation-set covariance tuning as our classical reference.

2.3 GPS Outage Bridging

Recent work has applied attention-based sequence models to GPS outage bridging in ground vehicle navigation, fusing learned velocity predictions into adaptive Kalman filters. These approaches retain the filter as a fusion target; GateIO removes the filter entirely during the outage and adds an explicit physics-motivated training constraint on low-yaw-rate windows.

2.4 Dataset

The MARS-LVIG dataset (Li et al., 2024) was collected at the University of Hong Kong and Beihang University using a DJI M300 RTK multirotor platform across four terrain types. The dataset was designed for LiDAR-visual-inertial SLAM evaluation. We used five of the 21 sequences from two terrain types; the remaining sequences were not available for download at the time of this study.

3. Dataset and Preprocessing

3.1 MARS-LVIG

We used five flight sequences from MARS-LVIG: gnss01, gnss02, and gnss03 from HK Airport terrain, and island_gnss02 and island_gnss03 from HK Island terrain. We used only IMU and GPS velocity channels — no visual, LiDAR, or barometric data. The dataset documentation (Li et al., 2024) should be consulted for exact sensor specifications; throughout this paper, IMU sample rate is 200 Hz as stated in the dataset paper, and GPS velocity is the RTK-derived velocity provided by the dataset. After preprocessing, the five sequences yield 16,544 total windows.

3.2 Windowing and Normalisation

Raw IMU data is segmented into 200-sample windows (1.0 s at 200 Hz) with a stride of 20 samples (0.1 s), giving a prediction rate of 10 Hz and integration timestep $DT = 0.1$ s. Each window is labelled with the GPS velocity at its centre sample. GPS velocity targets are normalised by the training-set inter-quartile range and median: $\hat{y}_{norm} = (y_{GPS} - median) / IQR$, with $IQR = [1.516, 7.985, 0.100]$ m/s. The gated loss targets the normalised last-known GPS velocity on low-yaw-rate windows, approximating a velocity-persistence constraint in normalised space.

Input windows have 14 channels: accelerometer xyz (channels 0–2), gyroscope xyz (3–5), attitude quaternion $[w, x, y, z]$ body-to-navigation frame, scalar-first convention (6–9), GPS velocity (10–12, zeroed during outage), and outage flag (13). IMU and GPS channels are normalised; quaternion channels are passed through unscaled.

3.3 Outage Simulation

For most sequences, GPS outages are simulated by zeroing channels 10–12 for 100 windows (10 s) starting at window 100 of each 300-window sequence. A three-window linear ramp transitions the outage flag at onset. The long-outage group (S53–S58) is evaluated with the same 10 s (100-window) outage mask, starting at window 100, as the other groups; these six sequences are analysed as a separate group because they exhibit the largest accumulated drift. A longer 30 s outage for this group is planned but is not used in the results reported here.

3.4 Sequence Groups and Dataset Split

Inspection of the validation sequences identified five groups. Straight-short (S0–S34, 35 sequences): level flight throughout the 10 s outage; constant-velocity mean drift 2.11 m. Straight-medium (S35–S40, 6 sequences): similar dynamics; 1.31 m. Turns (S41–S46, 6 sequences): coordinated manoeuvre during the outage, mean yaw rate 0.126 rad/s; 135.79 m constant-velocity drift. False-alarm (S47–S52, 6 sequences): near-constant GPS velocity with elevated lateral accelerometer amplitude from motor vibration; mean yaw rate 0.062 rad/s; 4.68 m constant-velocity drift. Long outage (S53–S58, 6 sequences): 30 s outage; 147.21 m constant-velocity drift. These groups were identified by inspecting the validation set before finalising the model and are therefore not fully independent of the evaluation.

The threshold $\tau = 0.10$ rad/s, derived from the validation-set distribution (mean 0.060 rad/s, std 0.089 rad/s), classifies all six S41–S46 sequences as turning and all six S47–S52 sequences as straight flight. This threshold is not fully independent of the evaluation.

The current split uses a chronological within-flight partition (80% training, 10% validation, 10% test). Post-evaluation analysis found that the 35 straight-short test sequences have mean headings concentrated

near $+10^\circ$ (northward), whereas the straight-short training sequences do not contain windows in that heading range; the training sequences from the same bags fly predominantly in approximately the opposite direction. This heading-distribution difference caused GateIO to give 30.39 m mean drift on those test sequences against 2.08 m for constant-velocity. A bag-level split that assigns complete flight sequences to separate partitions is in development.

4. Methodology

4.1 Problem Formulation

Let $\mathbf{x}_t \in \mathbb{R}^{(W \times C)}$ denote the input window at time t ($W = 200$, $C = 14$). Let $\mathbf{v}_t \in \mathbb{R}^3$ denote GPS velocity, available outside the outage and zeroed in the input during it. The model predicts $\hat{\mathbf{y}}_t$ in normalised GPS velocity space for each window. During the outage, position is estimated by integrating $(\hat{\mathbf{y}}_t \times \text{IQR} + \text{median}) \times \text{DT}$. The model uses IMU measurements (channels 0–9) together with the last available GPS velocity, retained via the TCN's temporal receptive field over pre-outage windows and via the velocity head's explicit $\mathbf{v}_{\text{prev}}$ pathway — it does not operate from inertial data alone.

4.2 Architecture

GateIO has four components. A window encoder maps each 200-sample window to a 48-dimensional token via three 1D convolutional layers (kernel sizes 7, 5, 3) with GroupNorm and GELU, followed by multi-head attention pooling against a learned query token. An outage positional encoding adds a sinusoidal embedding of elapsed outage time, resetting during GPS-aided windows. A causal TCN backbone processes the token sequence through two stacks of five dilated causal blocks (dilations 1, 2, 4, 8, 16). Each block contains two causal convolutions with kernel size 3; each such convolution with dilation d adds $(3-1) \times d = 2d$ causal positions to the receptive field. Each block therefore contributes $2 \times 2d = 4d$ positions (two convolutions). One stack contributes $4 \times (1+2+4+8+16) = 124$ additional positions. Two stacks contribute 248 additional positions; including the current token gives a total receptive field of 249 tokens (24.9 s at 10 Hz), sufficient to span any 10 s outage. ALiBi causal self-attention (Press et al., 2022) follows with four heads. A velocity head produces predictions through aided and outage branches, with the forwarded $\mathbf{v}_{\text{prev}}$ added to the outage-branch token before prediction.

Table 1. GateIO architecture. Total: $\sim 186,000$ trainable parameters.

Component	Description
Window Encoder	Three 1D conv layers ($k=7,5,3$, GroupNorm, GELU) + multi-head attention pooling $\rightarrow$ 48-dim token per 200-sample window (1.0 s of raw IMU)
Outage PE	Sinusoidal encoding of elapsed outage time; resets to zero during GPS-aided windows
Causal TCN	2 stacks $\times$ 5 dilated causal blocks (dil 1,2,4,8,16); each block has 2 conv layers ($k=3$). Total receptive field: $2 \times 4 \times (1+2+4+8+16) + 1 = 249$ tokens (24.9 s)
ALiBi Attention	4-head causal self-attention with linear position bias (Press et al., 2022); applied once after TCN
Velocity Head	Dual-branch (aided / outage); $\mathbf{v}_{\text{prev}}$ added to outage-branch token before prediction

4.3 Loss Function

$$L = L_{\text{data}} + \lambda_{\text{dr}} \cdot L_{\text{dr}} + \lambda_{\text{cvprior}} \cdot L_{\text{cvprior}} + \lambda_{\text{smooth}} \cdot L_{\text{smooth}}$$

L_{data} : Huber loss ($\delta=0.3$) between $\hat{y}_{t,\text{norm}}$ and $y_{t,\text{norm}}$ on aided windows. L_{dr} : same loss on outage windows, axis weights [1.0, 1.0, 3.0]. $\lambda_{\text{dr}} = 0.90$, $\lambda_{\text{cvprior}} = 0.50$, $\lambda_{\text{smooth}} = 0.001$. Axis weights apply to L_{dr} only.

4.4 Yaw-Rate-Gated Velocity-Persistence Loss

During windows with low yaw rate, horizontal velocity should remain close to its last known value. GateIO encodes this as a Huber loss between the normalised prediction and the normalised last-known GPS velocity, applied to outage windows where $|\omega_z| < \tau = 0.10$ rad/s:

$$L_{\text{cvprior}} = \text{Huber}(\hat{y}_{t,\text{norm}}, v_{\text{prev},\text{norm}}), \quad t \in S_{\text{outage}} \text{ where } |\omega_z| < \tau$$

where S_{outage} is the set of outage windows satisfying the gate condition. The axis weights [1.0, 1.0, 3.0] do not apply to L_{cvprior} . The gate switches off entirely on windows above τ , leaving L_{dr} to supervise predictions on those windows without interference from the persistence constraint.

The constraint is motivated by the observation that low yaw rate is often associated with near-constant horizontal velocity over short intervals. This relationship is particularly natural for approximately coordinated fixed-wing cruise, but we use it here as a data-driven motion prior rather than as a precise aerodynamic model.

4.5 Gate Configuration Ablation (R19–R22)

Ungated at $\lambda=0.10$ (R19): prediction and persistence losses conflict on turn windows, producing a $4\times$ turn regression. Gyro-gated at $\lambda=0.50$ (R20, current): disjoint window sets eliminate the conflict; best overall validation results. Gyro-gated at $\lambda=2.0$ (R21): turn performance degrades substantially (26.59 m on S41–S46 vs 4.73 m) because the higher weight dominates on low-yaw-rate transitional portions of turning sequences. Truth-gated at $\lambda=1.0$ (R22): adding a GPS-consistency condition to the training gate reduced S47–S52 coverage from 87.3% to 61.3%, creating a train/inference mismatch and raising false-alarm drift from 32.56 m to 53.27 m.

4.6 TCN Ablation Sensitivity

Zeroing all TCN weights post-training on the first ten validation sequences gives approximately $1\times$ drift increase under a prediction-only objective and $43.97\times$ under the gated loss. We interpret this as the gated loss creating a temporal retrieval task that the TCN is well suited to perform — reproducing v_{prev} over the outage requires retaining that value through the 249-token receptive field. This is a hypothesis consistent with the ablation pattern; confirming it would require gradient analysis and matched parameter-count controls.

The explicit v_{prev} -branch ablation — zeroing channels 10–12 of the input in the outage windows only, so that the velocity head's explicit shortcut receives zeros while historical GPS velocity in pre-outage windows remains accessible via the TCN — gives approximately $1\times$ drift increase. This indicates the model does not rely on the explicit shortcut; it accesses prior velocity through temporal context instead.

4.7 Training Protocol

AdamW, $\text{lr} = 8 \times 10^{-4}$, weight decay 10^{-4} , one-cycle schedule, batch size 16, up to 200 epochs, early stopping after 60 epochs without improvement in mean validation drift. Turn sequences are up-weighted by $8\times$ in the sampler. Temporal dropout (25%) randomly zeros IMU windows. A drift-based auxiliary loss is ramped in between epochs 30 and 70. All results are from single training runs; stability across seeds is unknown.

5. Experiments

5.1 Baselines

Constant velocity: Last GPS-aided velocity held constant for the outage duration.

LSTM: Identical to GateIO with the causal TCN and ALiBi attention replaced by a two-layer unidirectional LSTM (hidden 128, dropout 0.15). All other components and training settings are identical.

EKF: Six-state linear Kalman filter $[v_x, v_y, v_z, b_x, b_y, b_z]$. Body-frame accelerometer readings are rotated to the navigation frame via the attitude quaternion and corrected for gravity ($+9.81 \text{ m/s}^2$ on the navigation-frame z-axis). Q and R are tuned on the validation set via Nelder-Mead ($Q_v = 5.31 \times 10^{-5}$, $Q_{\text{bias}} = 3.88 \times 10^{-5}$, $R = 5.08 \times 10^{-6}$). Both the EKF parameters and GateIO hyperparameters are validation-selected; these are model-selection results.

5.2 Validation Results

Group	GateIO	LSTM	EKF	Const-v	N
Straight-short	0.63	0.64	9.51	2.11	35
Straight-medium	3.04	3.18	8.66	1.31	6
Turns ★	4.73	2.90	351.64	135.79	6
False-alarm	28.08	39.62	26.62	4.68	6
Long outage	26.59	27.09	293.10	147.21	6
Mean (m)	6.72	7.78	74.80	30.64	59
Median (m)	0.90	0.73	10.44	2.37	
90th pctl (m)	27.82	37.95	287.79	117.09	
% under 5 m	76.3%	79.7%	13.6%	74.6%	

Table 2. Per-group mean endpoint drift (m) on 59 validation sequences. ★ Primary group. Bold: lower of the two learned methods. All groups use the same 10 s outage mask (100 windows). All numbers are model-selection results on the within-flight split.

Both learned methods substantially outperform the baselines on the turn group; the LSTM achieves lower mean drift than GateIO (2.90 m vs 4.73 m). On the false-alarm group, both learned methods are worse than constant-velocity (28.08 m and 39.62 m vs 4.68 m). The EKF degrades on turns (351.64 m) and long outages (293.10 m), and its straight-segment performance (9.51 m vs 2.11 m constant-velocity) reflects the Nelder-Mead objective being dominated by the catastrophic groups.

5.3 Test Results

The heading-distribution bias described in Section 3.4 is most visible in the straight-short group: GateIO gives 30.39 m against 2.08 m for constant-velocity. On the turn group, both learned methods retain an advantage over constant-velocity (83.73 m and 68.48 m vs 598.18 m), but absolute errors are substantially higher than on validation. Aggregated over all 59 test sequences, the constant-velocity baseline attains a far lower median endpoint error (2.06 m versus 28.24 m for GateIO) and stays under 5 m on 84.7% of outages versus 0.0% for both learned methods, so GateIO's test-set advantage is confined to mean drift, which the turn group dominates.

Group	GateIO	LSTM	EKF	Const-v	N
Straight-short	30.39	35.26	11.24	2.08	35
Straight-medium	17.48	28.83	10.72	2.17	6
Turns ★	83.73	68.48	408.56	598.18	6
False-alarm	16.28	44.33	11.56	1.56	6
Long outage	40.65	70.56	75.45	5.19	6
Mean (m)	34.11	42.50	58.16	62.97	59
Median (m)	28.24	42.79	11.19	2.06	
90th pctl (m)	41.36	70.71	165.23	96.43	
% under 5 m	0.0%	0.0%	11.9%	84.7%	

Table 3. Per-group and aggregate endpoint drift (m) on 59 test sequences. Straight-flight results reflect heading-distribution bias (Section 3.4).

5.4 Diagnostic Results

Diagnostic	GateIO	LSTM	EKF	Note
TCN ablation — gated loss (×)	43.97	—	N/A	Single run, 10 val seqs
TCN ablation — pred-only (×)	~1.00	—	N/A	Little measurable effect
Explicit v_prev-branch ablation (×)	~1.00	~1.00	N/A	Outage windows only; pre-outage GPS intact

IMU temporal shuffle ($\times$)	1.07	—	N/A	Per-window features used
Gate coverage (straight)	84.5%	84.5%	N/A	At $\tau=0.10$ rad/s, val set

Table 4. Diagnostic results. Ablation ratios are from single runs on the first ten validation sequences. v_{prev} -branch ablation zeros channels 10–12 of the input in outage windows only; pre-outage GPS channels remain available via the TCN’s temporal context.

6. Discussion

6.1 EKF Bias Analysis

The EKF degrades substantially on turning sequences (351.64 m). For one representative sequence (S41), the pre-outage lateral accelerometer mean was -1.60 m/s² and the effective lateral acceleration implied by GPS velocity changes during the outage was $+2.83$ m/s², giving a discrepancy of 4.43 m/s². This discrepancy is consistent with the observed position error: 4.43 m/s² integrated over 10 s produces a velocity error of approximately 44 m/s, which propagates into a large position error. However, the 4.43 m/s² quantity conflates sensor bias with aircraft dynamics, gravity-compensation residuals, and GPS differentiation noise; we describe it as a bias-estimate discrepancy rather than a demonstrated sensor failure.

6.2 False-Alarm Group

Motor vibration on S47–S52 produces lateral accelerometer amplitudes overlapping with genuine turning manoeuvres. The gyroscope classifies these as straight flight (mean yaw rate 0.062 rad/s), and $L_{cvprior}$ applies the persistence constraint. Despite this, GateIO gives 28.08 m against 4.68 m for constant-velocity. The window encoder likely maps vibration-affected IMU patterns to tokens similar to those from turns, because both appeared in training with correlated velocity targets. The gate constrains the output but cannot override encoder representations.

6.3 Temporal Context Sensitivity

The TCN ablation ratio rises from approximately $1\times$ under a prediction-only objective to $43.97\times$ under the gated loss. The most natural interpretation is that the gated loss creates a temporal retrieval task that the 249-token receptive field enables. Without the gated loss, per-window features suffice and the backbone receives weak gradient. This remains a hypothesis; confirming it would require gradient flow analysis and matched parameter-count controls with repeated seeds.

7. Limitations and Future Work

Dataset split. All results use a within-flight chronological split. Validation results are model-selection results. A bag-level split is in development. On the held-out test split, GateIO’s straight-flight error rises to 30.4 m and the constant-velocity baseline is more accurate on the typical case (test median 2.06 m

versus 28.24 m for GateIO; 84.7% of outages under 5 m versus 0.0%); GateIO's test-set advantage is limited to mean drift and to avoiding the large turn-group errors of the classical baselines.

Platform. MARS-LVIG uses a DJI M300 RTK multirotor. The yaw-rate-velocity relationship is more precisely motivated for fixed-wing aircraft. Evaluation on a fixed-wing platform is needed to validate the kinematic motivation.

Gate derivation. $\tau = 0.10$ rad/s was derived from the validation set used for performance reporting. A soft or learned gate derived from independent data would strengthen the evaluation.

Prediction target. The current model predicts absolute GPS velocity; the gated loss approximates a velocity-persistence constraint. An explicit velocity-increment formulation is in development.

Single-run results. All results are from single training runs. Ablation ratios and group means should be verified across multiple seeds before drawing firm conclusions.

No uncertainty output. GateIO produces point estimates without covariance, which is necessary for integration into a complete navigation system.

Dataset scale. Five sequences from two terrain types may not represent other aircraft, altitudes, speeds, or vibration profiles.

8. Conclusion

We presented GateIO, a yaw-rate-gated causal TCN for UAV GPS-outage bridging, evaluated on five MARS-LVIG flight sequences. On the current within-flight validation split, GateIO reduces mean endpoint drift on turning sequences from 135.79 m (constant-velocity) to 4.73 m; the LSTM baseline achieves 2.90 m on the same group. The TCN shows $43.97\times$ ablation sensitivity under the gated loss but approximately $1\times$ under a prediction-only objective, consistent with the prior encouraging temporal look-back. The false-alarm group and the within-flight split design expose limitations requiring vibration-robust representations and bag-level evaluation. The prior is motivated by fixed-wing coordinated-flight kinematics but is evaluated here on a multirotor dataset; evaluation on a fixed-wing platform is a natural next step. The most immediate priorities are bag-level evaluation, repeated-seed training, and an explicit velocity-increment prediction target.

References

- Brossard, M., Barrau, A., and Bonnabel, S. (2020). AI-IMU dead-reckoning. *IEEE Transactions on Intelligent Vehicles*, 5(4):585–595. doi:10.1109/TIV.2020.2980758
- Chen, C., Lu, X., Markham, A., and Trigoni, N. (2018). IONet: learning to cure the curse of drift in inertial odometry. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence (AAAI'18)*, pp. 6468–6476.
- Cioffi, G., Bauersfeld, L., Kaufmann, E., and Scaramuzza, D. (2023). Learned inertial odometry for autonomous drone racing. *IEEE Robotics and Automation Letters*, 8(5):2684–2691.
- Herath, S., Yan, H., and Furukawa, Y. (2020). RoNIN: Robust neural inertial navigation in the wild: Benchmark, evaluations, and new methods. In *Proceedings of IEEE ICRA*.
- Li, H., Zou, Y., Chen, N., Lin, J., et al. (2024). MARS-LVIG dataset: A multi-sensor aerial robots SLAM dataset for LiDAR-visual-inertial-GNSS fusion. *International Journal of Robotics Research*, 43(8):1114–1127. doi:10.1177/02783649241227968
- Liang, W., Meng, S., Zhang, R., and Luo, Q. (2026). A physics-informed dual-branch LSTM network for UAV position and attitude estimation using inertial sensors. *Sensors*, 26(13):4287. doi:10.3390/s26134287
- Liu, W., Caruso, D., Ilg, E., Dong, J., Mourikis, A., Daniilidis, K., Kumar, V., and Engel, J. (2020). TLIO: Tight learned inertial odometry. *IEEE Robotics and Automation Letters*, 5(4):5653–5660. doi:10.1109/LRA.2020.3007421
- Press, O., Smith, N., and Lewis, M. (2022). Train short, test long: Attention with linear biases enables input length extrapolation. In *Proceedings of ICLR*.
- Qiu, Y., Xu, C., Chen, Y., Zhao, S., Geng, J., and Scherer, S. (2025). AirIO: Learning inertial odometry with enhanced IMU feature observability. arXiv:2501.15659.
- Rao, B., Kazemi, E., Ding, Y., Shila, D. M., Tucker, F. M., and Wang, L. (2022). CTIN: Robust contextual transformer network for inertial navigation. *Proceedings of the AAAI Conference on Artificial Intelligence*, 36(5):5413–5421. [Verify page range and DOI from AAAI record before submission]